

AYUSH KASHYAP

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Summary

AI and backend engineer who has shipped production **LLM systems** — cutting **API latency from 1.8s to under 600ms** and improving **RAG retrieval precision by 28%** at a construction-tech startup. Comfortable owning systems spanning **ETL pipelines**, **multi-agent orchestration**, **WASM-compiled engines**, and **reinforcement learning agents**.

Experience

Software Engineer · Constructure AI

Dec 2025 – Apr 2026 · Remote

- Architected **RESTful microservices** for LLM-driven construction software, reducing **p95 API latency from 1.8s to <600ms** via Redis caching, connection pooling, and async job queues.
- Built scalable **ETL pipelines** processing **500MB+ BIM/IFC datasets** for downstream LLM context generation and structured retrieval.
- Developed **multi-agent orchestration** with **LangGraph**, coordinating **4+ specialized agents** across planning, retrieval, and synthesis while reducing cross-agent error propagation by **40%**.
- Implemented **hybrid RAG retrieval** combining **pgvector semantic search** with structured SQL filtering, improving retrieval precision by **28%**.

Projects

GoChess Engine

[Demo](#)

- Implemented **Minimax Search** with **Alpha-Beta Pruning**, reducing effective game-tree size by **60%** and enabling depth-5 analysis in **less than 200ms**.
- Compiled the engine to **WebAssembly (WASM)** for near-native browser execution and integrated **Web Workers** achieving **2–3× throughput improvements**.
- Integrated **Zobrist Hashing**, **Transposition Tables** (**100K+ cached positions/session**), and **Quiescence Search**.

PyGit — Distributed Version Control Engine

[GitHub](#)

- Re-implemented core **Git internals** including **Blob**, **Tree**, and **Commit** objects with **content-addressable storage** and **SHA-1 persistence**.
- Supported repositories with **100+ commits** while maintaining **O(1)** object lookup efficiency.
- Built a CLI supporting **init**, **add**, **commit**, **log**, demonstrating understanding of **DAG-based version tracking**.

PacmanRL — Deep Reinforcement Learning Agent

[GitHub](#)

- Trained a **DQN agent** with experience replay achieving stable gameplay within **500 episodes**.
- Designed custom **reward shaping** reducing average episode length by **30%** and achieving **2× higher scores** than random baselines.
- Engineered an **11-dimensional game-state feature pipeline** enabling policy generalization across unseen maze layouts.

Technical Skills

Languages: Go, TypeScript, JavaScript, Python, C++, C#, Java

AI / ML: LangChain, LangGraph, RAG, pgvector, Pinecone, TensorFlow, DQN, OpenCV, Multi-Agent Systems, Prompt Engineering

Backend & APIs: Node.js, Express.js, REST APIs, GraphQL, Microservices, Redis, PostgreSQL, ETL Pipelines, Async Processing

Frontend & Web: React, Next.js, React Native, WebAssembly (WASM), Web Workers

DevOps & Infra: Git, Docker, CI/CD, Distributed Systems, Performance Profiling

Education

Punjab Engineering College (PEC), Chandigarh

B.Tech in Mechanical Engineering

2023 – 2027

CGPA: 8.28 / 10